NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Department of Computer Science and Engineering

MID SEMESTER EXAMINATION, July-Dec. 2025

ECE_UG/ECE_UG_DD/ECE_UG_VLSI: Semester-1

Course Name: Introduction to Computing — Course Code: CS011201

Maximum Time: 2 hours — Max. Marks: 30

Instruction:

1. Attempt all questions and assume any suitable data, if necessary.
3. The Marks, CO (Course Outcome) and BL (Bloom's Level) related to questions are mentioned.

CO1= Understanding, CO2= Apply, CO3= Analyze

<table>
<tr><td>1.</td><td>List and explain the parameters used in the classification of computers.</td><td>1+6=7</td><td>CO1</td></tr>
<tr><td>2.</td><td>Describe the steps involved in the execution of an instruction by the CPU.</td><td>5</td><td>CO2</td></tr>
<tr><td>3.</td><td>Why is star topology more reliable than bus topology?</td><td>3</td><td>CO1</td></tr>
<tr><td>4.</td><td>What will be the output of the following C code?
<table>
<tr><td>A.
#include <stdio.h>
int main()
{
int x = 1, y = 0, z = 3;
x > y ? printf("%d", z) : return z;
}</td><td>B.
#include <stdio.h>
int main()
{
int x = 0, y = 2, z = 3;
int a = x & y | z;
printf("%d", a);
}</td></tr>
<tr><td>C.
#include <stdio.h>
void main()
{
int x = 1, y = 0, z = 5;
int a = x && y || z++;
printf("%d", z);
}</td><td>D.
#include <stdio.h>
void main()
{
int x = 1, z = 3;
int y = x << 3;
printf(" %d\n", y);
}</td></tr>
<tr><td>E.
#include <stdio.h>
int main()
{
int a = 10, b = 5, c = 3;
b != !a;
c = !!a;
printf("%d\t%d", b, c);
}</td><td></td></tr>
</table></td><td>2X5=10</td><td>CO3</td></tr>
<tr><td>5.</td><td>Write a 'C' program to find greatest among three numbers.</td><td>5</td><td>CO3</td></tr>
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